

Assignment 2 ML spring camp

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1 Poisson distribution GLM

General form of exponential distribution

$$p(y|\eta) = b(y) \exp(\theta^T T(y) - a(\eta))$$

Poisson distribution

$$p(y|\lambda) = \frac{e^{-\lambda} \lambda^y}{y!}$$

$$p(y|\lambda) = \frac{1}{y!} \exp(y \ln(\lambda) - \lambda)$$

Comparing, it is evident that $T(y) = y$, $\eta = \ln \lambda$, $A(\eta) = \lambda = e^\eta$

Thus, expected value, $\mu = \frac{dA}{d\eta} = e^\eta = \lambda$ Hence, $g(\mu) = \ln \mu = \eta$, the canonical response function

2 Multilabel classification

Was too lazy for latex so there's pdfs, uploaded in same file